Face to Face: How the giants position themselves

AMD

Athlon XP

This is the desktop debut of the Palamino core. It can push core clock speed up to 1.6 GHz.

Athlon

Sporting 256 KB full speed cache, this thunderbird core processor can go up to 1.4 GHz.

Duron

Based on a scaled down version of the desktop core, this features a reduced L2 cache and a clock speed of up to 1.2 GHz.

INTEL

Pentium 4

It sports radical concepts such as Hyper Pipelined Technology which allow clock speed up to 2 GHz.

Pentium III

Using Intel's very own SSE instruction set, this chip had 256 KB of full speed Level 2 cache and could go up to 1 GHz.

Celeron

An improved version of Celeron, this offers a higher clock speed (1.2 GHz) and incorporates a 100 MHz Front Side Bus (FSB).

optimising multimedia applications. These optimisations have primarily concentrated on delivering smoother and richer content at a faster pace for multimedia. These instruction sets allow processing of large amounts of similar data (like video content) at once using one instruction. This significantly speeds up the processing.

Working with audio and video content is probably the most CPU-intensive task and hardly any other task can demand as much processor-power as video encoding. In the video encoding test, we encoded a standard MPEG video of 5 minutes into AVI at a bitrate of 900 Kbps. The scores logged in these tests are for small files but if you extrapolate them, the differences would be enormous when it comes to encoding an entire CD full of audio, or while downsampling video content that is in Gigabytes. The time taken may even run into hours.

If your work demands working with multimedia or audio-video content, then you should consider a processor that falls in the Elite range. The Athlon XP 1800+ was fastest here too, encoding the entire file in less than 100 seconds (99 seconds to be exact). The Pentium 4, 2 GHz took about 111 seconds to complete. Though the difference is only 12 seconds, if you extrapolate these numbers for an hour-long video, you will find out that the Pentium 4, 2 GHz takes slightly more than 2 minutes longer to complete the encoding process than the Athlon XP 1800+.

Even in the audio encoding test where we encoded a 50.3 MB WAV file to MP3 at 192 Kbps, we found a difference of over 5 seconds in the time logged by the Athlon XP 1800+ (20 seconds) and the Pentium 4, 2 GHz (25 seconds).

In the performance segment, the Athlon 1.4 GHz was the fastest by far, taking 108 seconds to complete the video encoding and 21 seconds for the audio encoding. In fact, it was faster even than the Pentium 4, 2 GHz here. The top Pentium III was quite slow in these tests, taking 149 seconds for video encoding and 35 seconds for audio; it was beaten by the similarly clocked Athlon 1.1 3 GHz by a fair margin too. The Athlon 1.4 GHz should serve you well if you are an amateur dabbling in multimedia and audio-video editing or want a PC that will give you a good output when viewing high-quality multimedia content.

In the value segment, the Duron 1.1 GHz took 141 seconds for the video encoding test as against the 164 seconds taken by the Celeron 1.2 GHz.

Professional applications and FPU strength

Users of software such as Maya 3D, AutoCAD and Photoshop have a constant complaint about the time it takes to render. If you ask them, then probably no speed is enough, and probably even Athlon XP 1800+ (which performed the best in these tests) would be slow for them.

All the processors were tested for image processing, ray tracing and intensive rendering tests. All of these tests primarily focus on the FPU strength of the processor.

Here we used the Photoshop filter test, POV Ray for the ray tracing test and SpecViewperf for the rendering tests. The Photoshop test applies 21 filters one after the other to an image and calculates the time taken for each filter to execute and complete.

POV Ray mathematically models the light rays as they bounce around a virtual world, reflecting and refracting. This can quite literally involve millions of floating point calculations and has proved to be an extremely stressful test. ➜

Fine-tuning Performance

So you've bought the latest speed demon and fail to understand why it's not living up to all the hype. Well, part of the problem lies in the way you configure your system. The first and the most important part of tweaking starts not from within Windows but from the BIOS of your system.

Here are few ways to tweak the BIOS to perfection:

1. Enable the Internal and External caches. The External cache in most cases points towards the L2 cache. This should never be disabled.

2. CAS/RAS latencies/DRAM cycle time: This setting affects the entire memory subsystem and the lower the latency, the better it is. A setting of 2 is ideal and will significantly improve performance but if your memory is incapable of handling this setting, then you will see random lockups and freezes.

3. PCI Dynamic Bursting, PCI-to-DRAM Prefetch, CPU-to-PCI write buffer: Make sure you have all of these enabled. All of these in combination will yield a sizeable amount of performance, especially in the CPU to PCI write buffer. When this feature is enabled, up to four data words can be written to the buffer to be queued to the PCI when it is ready to receive data. If this feature is disabled, the CPU can only write to the PCI bus directly and has to wait for the PCI bus to be ready to receive data. Enabling the buffer can drastically reduce the wait stages (idle cycles) of the CPU.

CAUTION: Before changing any settings make sure you write down all the original settings on a piece of paper. Do not rely on the BIOS fail safe and default settings, though they will bring back your BIOS to a factory setting. If you are technically inclined, you can download the latest current version of your BIOS (if available) from your motherboard manufacturer's Web site.